

Mathematical Models in Biotechnology

Jeffrey D. Varner

Robert Frederick Smith School of Chemical and Biomolecular Engineering,
Cornell University, Ithaca, NY, USA

Abstract

Predicting the intracellular flux distribution of a growing cell is a central problem in biotechnology, yet the internal fluxes, kinetic parameters, and metabolite concentrations that would determine it are, at genome scale, largely unmeasured, while the stoichiometry of the reaction network is fully known. This chapter develops flux balance analysis, the constraint-based method that turns stoichiometry and conservation into a linear program for the flux distribution, as an integrative framework in which the flux bounds, rather than the stoichiometric matrix, carry the biological information that narrows a prediction. The flux constraint is derived from an open mole balance on a growing culture rather than asserted as a steady state, a derivation that retains the growth-dilution term the textbook form discards and makes explicit the operating regimes, fed-batch and cell-free among them, where that term cannot be dropped. Each flux bound is then written as a product of dimensionless factors, one gateway apiece for thermodynamic reversibility, enzyme kinetics, gene expression, and regulation, so that supplying a thermodynamic estimate, a turnover number, an expression measurement, or a regulatory model amounts to opening the corresponding gateway. Two worked steady-state examples, a urea-cycle metabolic network constrained entirely from public databases and a feedback-inhibited pathway whose regulated throughput is recovered by opening the regulatory gateway against an independent kinetic truth, make the construction concrete, and two published systems, a dynamic cell-free reaction and a model-guided glycan-producing strain, show the gateways opened together at genome scale. Estimating a flux distribution from data and designing one by intervention emerge as the same operation on the bounds, run in opposite directions.

1 Introduction

Biotechnology depends on predicting and directing cellular metabolism. Metabolic engineering redirects flux toward a target product, bioprocess development tunes feeding and environment to sustain it, and strain design asks which genetic changes will move a phenotype in a wanted direction; each is, at bottom, a question about the intracellular flux distribution, the rates at which the reactions of metabolism actually run under a given condition [1]. That distribution is what a quantitative model is asked to predict, and it is also what is hardest to observe. A genome sequence and the stoichiometry of its annotated reactions are knowable in broad outline, though real reconstructions carry their own uncertainty in reaction membership, directionality, compartmentalization, and biomass composition; the internal fluxes, the kinetic parameters that would set them, and the metabolite concentrations that drive them are, at genome scale, far less constrained still. The problem is therefore to predict a metabolic phenotype from what can be known rather than from what a direct calculation would require.

Two broad strategies answer that problem. Mechanistic kinetic models write an explicit rate law for every reaction, from Michaelis–Menten saturation [2] through mass action to the power-law

formalism of biochemical systems theory [3, 4], and integrate the resulting equations forward in time; they resolve the dynamics in full but demand a rate constant, an affinity, or a kinetic order for every step, parameters that are rarely available beyond a handful of well-studied pathways. Constraint-based models make the opposite trade. They ask only for the stoichiometry of the network and impose conservation of mass at steady state, and in doing so they scale to genome-wide reconstructions of thousands of reactions [1, 5]. Flux balance analysis is the constraint-based workhorse: it casts phenotype prediction as a linear program over the flux vector [6, 7]. Conservation alone, however, does not determine a flux distribution; it fixes a whole subspace of them, and what selects a prediction from that subspace is the box of upper and lower bounds placed on each reaction. Flux bounds are a major interface through which condition-specific thermodynamic, kinetic, expression, and regulatory information enters a constraint-based model, and the question this chapter takes up is how it enters: through which bound does a thermodynamic estimate, a turnover number, an expression measurement, or a regulatory signal actually change a flux prediction?

This chapter develops flux balance analysis as that integrative framework, organized around the bounds as the gateway through which every class of biological information enters a flux estimate. The flux constraint itself is derived not by asserting steady state but from an open mole balance on a growing culture, a derivation that keeps the growth-dilution term the textbook form discards and writes the constraint as $\mathbf{S}\hat{\mathbf{v}} = \mu\mathbf{x}$ rather than $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$. A single bound is then read as a product of factors, one gateway each for thermodynamic reversibility, enzyme kinetics, gene expression, and regulation, so that opening a gateway means supplying the data or the model that sets its factor. Two worked steady-state examples make the construction concrete: a urea-cycle metabolic network whose flux distribution is pinned by thermodynamic and kinetic data drawn entirely from public databases, and a feedback-inhibited pathway whose regulatory truth, generated independently from a kinetic model, is recovered by opening the regulatory gateway on a single bound. Two reported systems then show the same gateways opened together at a scale no hand calculation could reach, one estimating fluxes in a dynamic cell-free reaction [8] and one designing a glycan-producing strain by model-guided gene deletion [9]. Read across these cases, estimating a flux distribution from data and designing one by intervention evaluate the bounds the same way once an intervention is chosen; design differs in also requiring a combinatorial search, over candidate deletions or expression changes, for which bounds to move [10].

2 From Mole Balances to the Flux Constraint

Flux balance analysis (FBA) rests on an algebraic constraint rather than a differential equation, and that constraint is a mole balance in disguise. Writing the balance out, instead of asserting metabolic steady state by fiat, makes explicit what is assumed away when the constraint is reduced to its familiar textbook form, and which operating conditions the reduction cannot cover. Consider a well-mixed bioreactor of volume V holding a growing cell culture. The culture exchanges mass with its surroundings through a set of physical streams s (feed, harvest, gas sparging) and hosts an intracellular reaction network with n fluxes $\hat{v}_j$, $j = 1, \dots, n$, acting on m metabolite species. For a species i present at concentration C_i , an open mole balance over the whole culture accounts for material entering and leaving through the streams together with material produced or consumed by the reaction network:

$$\sum_s d_s C_{i,s} \dot{V}_s + \sum_j \sigma_{ij} \hat{v}_j V = \frac{d}{dt}(C_i V), \quad (1)$$

where $d_s = +1$ for streams entering the reactor and $d_s = -1$ for streams leaving it, $C_{i,s}$ is the concentration of species i in stream s , $\dot{V}_s$ is the volumetric flow rate of that stream, and σ_{ij} is the stoichiometric coefficient of species i in reaction j , the (i, j) entry of the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m \times n}$. Equation (1) is exact: it presumes nothing about growth rate, feeding strategy, or steady state, and it is the starting point for the reduction carried out below.

Two steps turn Equation (1) into a statement about fluxes alone. First, write the reactor volume in specific units, $V = B\bar{V}$, with B the total biomass in the reactor and $\bar{V}$ the culture volume per unit biomass; this ties the extensive balance to the biomass that is actually growing, and the specific growth rate follows as $\mu \equiv (1/B) dB/dt$. Second, expand the accumulation term by the product rule,

$$\frac{d}{dt}(C_i V) = \frac{d}{dt}(C_i B \bar{V}) = B \bar{V} \frac{dC_i}{dt} + C_i \bar{V} \frac{dB}{dt} + C_i B \frac{d\bar{V}}{dt}. \quad (2)$$

Dividing Equation (1) through by $V = B\bar{V}$ and substituting Equation (2) isolates the intensive balance for species i ,

$$\frac{1}{B\bar{V}} \sum_s d_s C_{i,s} \dot{V}_s + \sum_j \sigma_{ij} \hat{v}_j = \frac{dC_i}{dt} + \mu C_i + \frac{C_i}{\bar{V}} \frac{d\bar{V}}{dt}, \quad (3)$$

where the middle term on the right, μC_i with $\mu = (1/B) dB/dt$, is the growth-dilution term produced by the product rule. Three assumptions each remove one of the remaining terms. If the intracellular pool of species i is at pseudo-steady state ($dC_i/dt = 0$, the usual separation-of-timescales argument, since metabolite turnover on the order of seconds to minutes is fast relative to a doubling time of hours), if the specific culture volume is constant ($d\bar{V}/dt = 0$, so the working volume tracks the growing biomass rather than drifting relative to it), and if species i crosses no physical stream (the sum over s vanishes for this species), Equation (3) collapses to

$$C_i \mu = \sum_j \sigma_{ij} \hat{v}_j \quad \Longleftrightarrow \quad \boxed{\mathbf{S}\hat{\mathbf{v}} = \mu \mathbf{x}}, \quad (4)$$

where $\mathbf{x} = (C_1, \dots, C_m)^\top$ collects the intracellular concentrations and $\hat{\mathbf{v}} \in \mathbb{R}^n$ collects the fluxes. Equation (4) is the honest flux constraint: net metabolic production of every intracellular species is balanced not against zero, but against the rate at which that species is diluted by its own growing culture.

The form used throughout the constraint-based modeling literature is obtained by discarding the right-hand side of Equation (4) outright:

$$\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}. \quad (5)$$

Equation (5) is a labeled special case of Equation (4), not an independent starting point [6]. It is a good approximation exactly when the two conditions that justify dropping $\mu \mathbf{x}$ hold together: growth is slow relative to metabolic turnover, and intracellular pools are dilute enough that $\mu \mathbf{x}$ stays small in absolute terms even when μ itself is not negligible. Both conditions can fail in operating regimes of real interest. In fed-batch culture the feed is metered to increase the working volume over the course of a run, so the specific volume drifts ($d\bar{V}/dt \neq 0$); the assumption used to reach Equation (4) no longer holds, and the discarded term carries information about the feed schedule that Equation (5) cannot see. In cell-free systems there is no growing cell to dilute into ($\mu = 0$), yet the reaction mixture is manifestly not at steady state: transcript, protein, and metabolite pools accumulate and decay over the course of a batch reaction ($d\mathbf{x}/dt \neq 0$), so the pseudo-steady-state

assumption fails from the other direction. Neither shortfall is a corner case; a cell-free capstone example later in this chapter returns to exactly this setting, where the term Equation (5) discards is not a formal nuisance but a quantity under direct experimental measurement.

Equation (4) and its special case Equation (5) are, as written, statements about a closed system: every species that appears has its production and consumption accounted for entirely by the internal network $\mathbf{S}\hat{\mathbf{v}}$, with the physical stream terms of Equation (1) already discarded along the way. Real metabolism is open, since cells take up substrates and secrete products across the boundary of whatever system is being modeled. Constraint-based reconstructions restore that openness without reintroducing the stream terms explicitly: hypothetical exchange, or boundary, reactions are added to the network itself, one pseudo-reaction $\emptyset \rightleftharpoons M_i$ for each boundary metabolite M_i , entering $\mathbf{S}$ as an isolated nonzero column with no reaction partner [1]. Flux through an exchange reaction enters Equation (4) or Equation (5) exactly as any metabolic flux does, but it is now interpretable as uptake or secretion across the system boundary rather than as a step of intracellular chemistry. A closed algebraic statement can thereby represent a genuinely open system: openness becomes entirely a matter of which reactions are admitted into $\mathbf{S}$ and how their bounds are set, the question taken up next.

3 The Linear Program and Its Geometry

The honest constraint of Equation (4), together with its steady-state special case Equation (5), is a statement of conservation and nothing more: it does not by itself pick out any particular flux distribution among all those consistent with it. Turning conservation into a prediction requires two further ingredients. The first is a box of flux bounds, $\boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}$, one lower and one upper bound per reaction, encoding thermodynamic reversibility ($\ell_j < 0$ permitted), enzyme capacity, and substrate availability; how those bounds are set is taken up next. The second is an objective, a linear functional $\mathbf{c}^\top \hat{\mathbf{v}}$ of the flux vector that states what the cell, or the modeler, is presumed to optimize. With both in hand the mole-balance constraint becomes a linear program:

$$\begin{aligned} \max_{\hat{\mathbf{v}} \in \mathbb{R}^n} \quad & \mathbf{c}^\top \hat{\mathbf{v}} \\ \text{subject to} \quad & \mathbf{S}\hat{\mathbf{v}} = \mathbf{0} \quad (\text{or } \mu\mathbf{x}), \\ & \boldsymbol{\ell} \leq \hat{\mathbf{v}} \leq \mathbf{u}. \end{aligned} \tag{6}$$

Equation (6) is flux balance analysis in its complete, textbook form [6]. The objective vector $\mathbf{c}$ is a choice, not a derived quantity: setting $c_j = 1$ at the index of a biomass pseudo-reaction, one that drains precursors in the ratios required to build new cell mass, casts the LP as growth-rate maximization, the default assumption for a cell under selection pressure to divide. Setting $c_j = 1$ instead at the index of a product-export exchange reaction casts the same LP as yield maximization for a chosen metabolite, the framing relevant to a bioprocess engineer optimizing a production strain rather than modeling wild-type physiology. Nothing in Equation (6) distinguishes these cases mechanistically; the constraint set is identical, and only $\mathbf{c}$ changes.

The linear-programming machinery is necessary, rather than a matter of convenience, because the constraint set alone is almost always underdetermined. A genome-scale reconstruction lists every reaction annotated to a genome, and the reaction count n typically exceeds the metabolite count m , because a single metabolite typically participates in several reactions while a single reaction typically touches only a handful of metabolites. The stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{m \times n}$ therefore has many more columns than rows, and its null space $\mathcal{N}(\mathbf{S}) = \{\hat{\mathbf{v}} \in \mathbb{R}^n : \mathbf{S}\hat{\mathbf{v}} = \mathbf{0}\}$ is nontrivial: a whole subspace, not a point, satisfies the steady-state constraint. Every flux vector

in $\mathcal{N}(\mathbf{S})$ conserves mass at every internal node exactly as well as every other; conservation alone cannot distinguish among them. The flux bounds cut this unbounded subspace down to a bounded polytope, and the objective in Equation (6) selects a single vertex, or a face, of that polytope, but the underlying indeterminacy, that most of the space allowed by stoichiometry is not resolved by stoichiometry, does not go away merely because an optimum has been found.

The singular value decomposition makes the geometry of that indeterminacy explicit. Writing $\mathbf{S} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, with $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$ orthogonal and $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ diagonal with nonnegative entries, the columns of $\mathbf{V}$ associated with zero singular values span $\mathcal{N}(\mathbf{S})$, the right null space: every feasible steady-state flux mode is a linear combination of these columns, and reading Equation (6) through this lens, the LP is a search over coefficients of that combination rather than over the raw flux vector. The columns of $\mathbf{U}$ associated with zero singular values span the left null space of $\mathbf{S}$, equivalently the null space of $\mathbf{S}^\top$; a vector $\mathbf{y}$ in this space defines a conserved moiety, a linear combination $\mathbf{y}^\top \mathbf{x}$ of metabolite concentrations, such as a bound cofactor pair sharing a fixed pool total, that is invariant under the reaction network regardless of which flux vector in $\mathcal{N}(\mathbf{S})$ is realized. For a genome-scale reconstruction the rank of $\mathbf{S}$ is smaller than the reaction count, $\text{rank}(\mathbf{S}) \ll n$, so the right null space, and with it the space of flux distributions the constraints alone cannot distinguish, is large [1]. That indeterminacy can be read reaction by reaction. Because an LP optimum need not be unique, and because many reactions can vary freely without changing the objective value at all, flux variability analysis reports a range rather than a point for each flux: with the objective fixed at its optimal value Z^* (or at a specified fraction of it), each v_j is in turn minimized and maximized subject to Equation (6)’s constraints [7]. A reaction whose minimum and maximum coincide is pinned by the network topology and bounds at the optimal objective value used for that calculation, while a reaction with a wide range is left essentially free; a different objective can produce a different feasible face and a different range for the same reaction, and the pattern of which fluxes fall into which category is exactly the information that a single optimal flux vector, taken alone, cannot report.

4 Bounds as Gateways

Conservation alone leaves the flux distribution underdetermined: the mole-balance constraint fixes a subspace, not a point, and the box of bounds together with the objective is what narrows that subspace to a prediction. The objective is a modeling choice. The bounds are where biology enters, and they enter not as numbers to be looked up but as a model in their own right. Writing a single upper bound as a product of factors makes that model visible. For reaction j ,

$$-\delta_j \left[V_{\max,j}^\circ (e/e^\circ) \theta_j(\cdot) f_j(\cdot) \right] \leq \hat{v}_j \leq V_{\max,j}^\circ (e/e^\circ) \theta_j(\cdot) f_j(\cdot), \quad (7)$$

where $V_{\max,j}^\circ$ sets a reference capacity carrying the units of flux and the remaining factors are dimensionless corrections that scale that capacity up or down. Each factor is a gateway for a distinct class of biological information: δ_j admits thermodynamics, $V_{\max,j}^\circ$ and f_j admit enzyme kinetics, (e/e°) admits gene expression, and θ_j admits regulation. A bound is therefore an honest ledger of what is known about a reaction. With no information beyond a capacity scale the correction factors are set to unity, and each is reopened, one gateway at a time, as measurements or mechanistic models become available. This is also where the underdetermined polytope is finally narrowed: not by conservation but by the bounds, and every gateway that closes tightens that polytope with information of a particular kind, so that tracing a prediction back to its cause is a matter of asking which factor in Equation (7) was responsible.

The thermodynamic gateway is the switch δ_j , which decides whether the reaction may carry flux in the reverse direction at all. It takes one of two values,

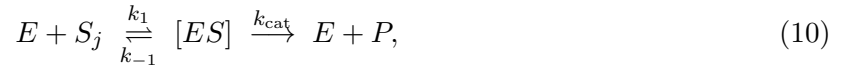
$$\delta_j = \begin{cases} 1, & \text{reaction } j \text{ reversible,} \\ 0, & \text{reaction } j \text{ irreversible,} \end{cases} \quad (8)$$

so that $\delta_j = 1$ leaves the lower bound of Equation (7) negative and the reaction free to run backward, while $\delta_j = 0$ collapses the lower bound to zero and pins the reaction to the forward direction. The assignment follows the sign of $\Delta G_j^\circ - \Delta G_j^*$, the standard free energy of reaction measured against a threshold ΔG_j^* meant to represent the physiological concentration range, or equivalently a cutoff on the equilibrium constant $K_{\text{eq},j}$: a reaction whose driving force cannot plausibly change sign anywhere in that range is declared irreversible, and its reverse flux is forbidden. This is a heuristic directionality rule, a stand-in for propagating a stated concentration range through $\Delta G = \Delta G^\circ + RT \ln Q$ and checking directly whether the sign can flip, not a substitute for that calculation where the range is actually known. Standard free energies for this test are drawn from group- and component-contribution estimates, for which eQuilibrator provides genome-scale coverage [11].

When the reaction is allowed to proceed, its capacity is set by enzyme kinetics, and this is the role of the reference scale $V_{\text{max},j}^\circ$ together with the saturation factor f_j . The scale is a maximal velocity, the product of a turnover number and a characteristic enzyme abundance,

$$V_{\text{max},j}^\circ = k_{\text{cat},j}^\circ e^\circ, \quad (9)$$

where $k_{\text{cat},j}^\circ$ is the catalytic rate constant of the enzyme catalyzing reaction j and e° is a reference enzyme concentration. Turnover numbers are tabulated per enzyme and substrate in BRENDA [12], and a representative e° can be read from compilations of cellular abundances such as BioNumbers [13]. The scale $V_{\text{max},j}^\circ$ is the flux the enzyme would sustain at reference abundance and saturating substrate, and the fraction of that ceiling actually reached at the prevailing substrate concentration is the saturation factor. That factor follows from the enzyme mechanism itself. A single-substrate enzyme binds substrate reversibly and turns it over to product,



with $[ES]$ the enzyme-substrate complex. Holding the complex at quasi-steady state, on the argument that it equilibrates fast relative to the depletion of substrate,

$$\frac{d[ES]}{dt} = k_1[E][S_j] - (k_{-1} + k_{\text{cat}})[ES] = 0, \quad (11)$$

and eliminating the free enzyme through the conservation of total enzyme $[E]_T = [E] + [ES]$, gives the complex in terms of the substrate alone,

$$[ES] = \frac{[E]_T [S_j]}{K_{M,j} + [S_j]}, \quad K_{M,j} \equiv \frac{k_{-1} + k_{\text{cat}}}{k_1}. \quad (12)$$

The reaction velocity is the rate of the turnover step, $v_j = k_{\text{cat}}[ES]$, so writing the capacity as $V_{\text{max},j} = k_{\text{cat}}[E]_T$ recovers the Michaelis–Menten rate law [2], and the saturation factor is its normalization to that capacity,

$$v_j = V_{\text{max},j} \frac{[S_j]}{K_{M,j} + [S_j]} \implies f_j([S_j]) = \frac{v_j}{V_{\text{max},j}} = \frac{[S_j]}{K_{M,j} + [S_j]} \in [0, 1], \quad (13)$$

with $[S_j]$ the substrate concentration and $K_{M,j}$ the Michaelis constant. The saturation factor f_j interpolates between an empty gateway ($f_j \rightarrow 0$ as substrate is exhausted) and a fully open one ($f_j \rightarrow 1$ at saturation), and it is the same hyperbolic law that governs single-enzyme assays. Because f_j depends on the substrate concentration, opening this gateway makes the bound a function of the metabolic state rather than a fixed number, the point at which a static flux balance begins to acquire the state-dependence that a dynamic treatment carries to its conclusion.

The scale $V_{\max,j}^\circ$ was fixed at a reference enzyme abundance e° , but the abundance actually present is itself a variable set by gene expression, and the ratio (e/e°) is the gateway through which that variability enters. To treat abundance as a modeled quantity rather than a fixed parameter, write balances for the transcript m_j and the protein p_j of the gene encoding the enzyme,

$$\dot{m}_j = r_{X,j} u_j - (\theta_{m,j} + \mu) m_j + \lambda_j, \quad (14)$$

$$\dot{p}_j = r_{L,j} w_j m_j - (\theta_{p,j} + \mu) p_j, \quad (15)$$

where $r_{X,j}$ and $r_{L,j}$ are transcription and translation rate capacities, u_j and w_j are dimensionless control variables that gate those capacities, $\theta_{m,j}$ and $\theta_{p,j}$ are first-order degradation rate constants for message and protein, and λ_j is a basal, or leak, source of transcript. The loss term in each balance is not the degradation constant alone but the sum $(\theta_{m,j} + \mu)$ or $(\theta_{p,j} + \mu)$: every intracellular species is removed both by its own turnover and by dilution into the growing culture at the specific growth rate μ . That μ is not a new parameter. It is the very growth-dilution term retained in the honest constraint, Equation (4), carried through consistently: an enzyme is diluted by growth exactly as a metabolite is, and the same μ that balances metabolic production in the flux constraint sets the residence time of the transcripts and proteins that fix the flux bounds. Holding the balances at steady state gives closed-form abundances,

$$m_j^* = \frac{r_{X,j} u_j + \lambda_j}{\theta_{m,j} + \mu}, \quad p_j^* = \frac{r_{L,j} w_j m_j^*}{\theta_{p,j} + \mu}, \quad (e/e^\circ) = \frac{p_j^*}{e^\circ}, \quad (16)$$

and the protein level p_j^* is mapped onto the abundance of the catalytic activity for reaction j through the gene-protein-reaction (GPR) rules of the reconstruction, which encode whether an activity requires a single gene product, several assembled into a complex, or any one of several interchangeable isozymes. Through Equation (16) the expression gateway couples the flux ceiling to the entire upstream apparatus of transcription and translation, with μ standing in every denominator as the visible signature of the growth-aware bookkeeping begun in the mole-balance derivation. The ratio (e/e°) is also the natural port for measured data: a transcriptomic or proteomic profile enters here, rescaling the capacity of each reaction by the abundance actually observed under a given condition and turning one reconstruction into a family of condition-specific models.

The factor θ_j is a control function, a dimensionless number in $[0, 1]$ reporting the fraction of a reaction's capacity switched on by the regulatory state of the cell, and the same machinery that sets it supplies the control variables u_j , w_j , and the leak λ_j left unspecified in the expression balances. A regulated element, a promoter or an allosteric enzyme, visits a discrete set of microstates s , and the fraction of time spent in each follows a Boltzmann distribution over the state energies,

$$p_s = \frac{f_s e^{-\beta \epsilon_s}}{Z}, \quad Z = \sum_s f_s e^{-\beta \epsilon_s}, \quad (17)$$

where ϵ_s is the energy of microstate s , $\beta = 1/(k_B T)$, Z is the partition function that normalizes the distribution, and f_s is a Hill-type function of the effector that stabilizes state s , $f_s = x^n/(K^n + x^n)$ in an activating effector concentration x . Summing the state probabilities over the microstates that

permit activity, and splitting that set into states reached through the regulated input $\mathcal{A}$ and states occupied without it $\mathcal{B}$, gives the total occupancy together with its basal part,

$$\bar{u} = \sum_{s \in \mathcal{A}} p_s + \sum_{s \in \mathcal{B}} p_s = u + u^\dagger, \quad \lambda \equiv r_X u^\dagger. \quad (18)$$

The regulated occupancy u is the control that appears in Equation (14), the basal occupancy $u^\dagger$ is exactly the leak source λ of that same balance, and the normalized activity $\bar{u}$ is the control function θ_j entering the bound: the regulatory and expression gateways are two views of one coupled process rather than independent knobs. This partition-function control function, Equations (17) and (18), is not a phenomenological Hill curve grafted onto a rate law. It is the effective biophysical model of Adhikari and coworkers [14], one object in which the Boltzmann weighting over states and the control function it produces are derived from the same statistical-mechanical accounting rather than assembled from two separate parts.

Every gateway just opened demands data that is often unavailable: thermodynamic estimates, turnover numbers, substrate concentrations, expression measurements, and regulatory parameters. When that data is scarce the correction factors are set to their information-free defaults, $f_j \approx 1$, $(e/e^\circ) \approx 1$, and $\theta_j \approx 1$, and the general bound collapses to a two-parameter box,

$$-\delta_j V_{\max,j}^\circ \leq \hat{v}_j \leq V_{\max,j}^\circ. \quad (19)$$

Equation (19) retains only the thermodynamic gate and the capacity scale, and it is exactly the reversible or irreversible $\pm V_{\max}$ box used throughout genome-scale flux balance analysis. It is the tractable baseline, the honest statement of a bound when nothing beyond a capacity estimate is known, and Equation (7) records precisely which assumptions that baseline makes and which gateway must be reopened to relax each of them.

Each gateway just described has a named, established predecessor in the constraint-based modeling literature, and this chapter’s contribution is the unifying notation and workflow that connects them, not the invention of any individual constraint. The thermodynamic gateway δ_j is the central device of thermodynamics-based flux analysis, which propagates standard free energies and metabolite activity ranges through the whole network to rule out infeasible flux and concentration profiles together [15]. The kinetic gateway, $V_{\max,j}^\circ$ built from a turnover number and an enzyme abundance together with the saturation factor f_j , is the organizing idea behind enzyme-constrained flux balance analysis, most fully realized in the GECKO framework, which folds a proteome-wide capacity constraint into a genome-scale reconstruction [16]. The expression gateway (e/e°) , transcript and protein balances entered as a modeled quantity rather than a fixed parameter, is the metabolism-and-expression (ME) model construction, which couples the transcription and translation machinery directly to the metabolic network at genome scale [17]. The regulatory gateway θ_j has two established lines of descent: regulatory flux balance analysis, which gates reactions on and off with Boolean rules derived from a transcriptional regulatory network [18], and PROM, which replaces the Boolean rule with a continuous, data-derived probability of an interaction being active [19], the same move from a binary switch to a graded control fraction that the partition-function construction of Equations (17) and (18) makes explicit. What the gateway factorization of Equation (7) adds is not a fifth constraint class but a common accounting of where each of these four already-established devices sits inside one bound, so that a single notation and a single worked derivation carry across all four rather than requiring a separate formalism, and a separate paper, for each.

The expression and regulatory gateways lumped transcription and translation into aggregate capacities $r_{X,j}$ and $r_{L,j}$, but these processes can themselves be resolved to the level of the sequences

they read. Transcription of a gene and translation of its protein consume nucleotides, amino acids, and energy in stoichiometric ratios fixed by the gene and protein sequences, and those precursor demands can be computed directly from the sequences and entered into $\mathbf{S}$ as reactions in their own right [5], with the precursors supplied by exchange reactions across the system boundary. Expression then becomes flux-bearing chemistry inside the same linear program as metabolism, a construction carried out for cell-free protein synthesis by Vilkhovoy and coworkers [20], and it is the mechanism behind the cell-free capstone taken up later in this chapter.

5 Worked Example: Urea-Cycle Metabolism

The general bound of Equation (7) carries four gateways, and a genome-scale reconstruction may in principle open all four at once. The urea cycle in HL-60 cells, parameterized here with cross-organism public data rather than HL-60-specific measurements, is small enough to make the opposite case concretely: two gateways, thermodynamic and kinetic, filled entirely from public databases, already pin down a sharp prediction without a single measurement taken on the cells themselves. The network retains the same steady-state constraint $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$, Equation (5), and the same linear program, Equation (6), established earlier. What changes from one application to the next is only the content poured into $\mathbf{S}$, ℓ , $\mathbf{u}$, and $\mathbf{c}$.

The reconstruction couples the four canonical urea-cycle enzymes to a fifth, competing branch and to the extracellular environment (left panel, Fig. 1). Argininosuccinate synthetase (v_1 , EC 6.3.4.5) condenses citrulline with aspartate and ATP to argininosuccinate, releasing AMP and diphosphate; argininosuccinate lyase (v_2 , EC 4.3.2.1) cleaves that intermediate to arginine and fumarate; arginase I (v_3 , EC 3.5.3.1) hydrolyzes arginine to ornithine and urea; and ornithine transcarbamylase (v_4 , EC 2.1.3.3) recombines ornithine with carbamoyl phosphate to regenerate citrulline, closing the cycle. A fifth reaction, nitric oxide synthase (v_5 , EC 1.14.13.39), draws on the same arginine pool consumed by v_3 and oxidizes it with NADPH and oxygen to nitric oxide and citrulline, diverting flux away from urea production toward signaling chemistry. These five enzymatic reactions act on eighteen metabolites. Fourteen of the eighteen, carbamoyl phosphate, aspartate, ATP, water, oxygen, NADPH, and protons on the uptake side, urea, fumarate, AMP, diphosphate, orthophosphate, NADP^+ , and nitric oxide on the secretion side, cross the system boundary through fourteen exchange reactions, b_1 through b_{14} , written in the standard secretion-positive convention $M_i \rightarrow \emptyset$, a positive flux denoting secretion and a negative flux uptake. The remaining four, arginine, citrulline, ornithine, and argininosuccinate, are pure cycle intermediates with no exchange reaction of their own: each is produced and consumed entirely inside the network. Collecting five enzymatic and fourteen exchange fluxes against eighteen metabolite balances gives $\mathbf{S} \in \mathbb{R}^{18 \times 19}$, $m = 18$ intracellular balances and $n = 19$ reactions.

The thermodynamic gateway of Equation (8) was opened first. Standard Gibbs free energies of reaction for all five enzymatic steps were evaluated by eQuilibrator [11]: $\Delta G_{v_1}^\circ = -4.3$, $\Delta G_{v_2}^\circ = -5.5$, $\Delta G_{v_3}^\circ = -51.0$, $\Delta G_{v_4}^\circ = -30.3$, and $\Delta G_{v_5}^\circ = -1220$ kJ/mol. The threshold ΔG_j^* of Equation (8) was fixed here at -10 kJ/mol, a heuristic cutoff rather than one derived from a shown physiological concentration range: a reaction whose standard free energy falls below that cutoff is declared irreversible and its reverse flux forbidden ($\delta_j = 0$), while a reaction whose free energy sits above it is left reversible ($\delta_j = 1$). Only v_1 and v_2 cleared the threshold and remained reversible; v_3 , v_4 , and v_5 fell below it, v_5 the furthest at -1220 kJ/mol, and were pinned irreversible. Every exchange reaction was left unconstrained by this test and kept a wide default bound in both directions, since the direction of nutrient uptake or product secretion is a matter of medium composition rather than of standard free energy. The kinetic gateway of Equation (9)

was opened next. Turnover numbers for the five enzymes were taken from BRENDA [12] and combined with a common reference enzyme abundance $e^\circ = 0.01$ mmol/gDW, a characteristic order of magnitude for intracellular enzyme concentration drawn from compilations such as BioNumbers [13]. Argininosuccinate synthetase and nitric oxide synthase share $k_{\text{cat}}^\circ = 10.0$ s⁻¹, giving, after converting the per-second turnover number to the per-hour basis the rest of the chapter uses, $V_{\text{max},v_1}^\circ = V_{\text{max},v_5}^\circ = 360$ mmol gDW⁻¹ h⁻¹; arginase I has $k_{\text{cat}}^\circ = 190.0$ s⁻¹, giving $V_{\text{max},v_3}^\circ = 6840$ mmol gDW⁻¹ h⁻¹; and ornithine transcarbamylase has $k_{\text{cat}}^\circ = 410.0$ s⁻¹, giving $V_{\text{max},v_4}^\circ = 14760$ mmol gDW⁻¹ h⁻¹. Argininosuccinate lyase is the outlier: $k_{\text{cat},v_2}^\circ = 3.28$ s⁻¹, an order of magnitude below the other four turnover numbers, gives $V_{\text{max},v_2}^\circ = 118.08$ mmol gDW⁻¹ h⁻¹, the smallest of the five capacities by more than threefold. Every exchange reaction was again left with a wide capacity bound, since a boundary reaction represents bulk transport rather than a single characterized enzyme.

With the thermodynamic and kinetic gateways open and the other two left shut, the general bound of Equation (7) collapses to the two-parameter box of Equation (19) for every enzymatic reaction: no substrate-concentration data enters the model, so the saturation factor is held at its information-free default $f_j = 1$; enzyme abundance is fixed at the reference value itself, so $(e/e^\circ) = 1$; and no regulatory information is imposed, so $\theta_j = 1$. The bound used in the linear program is therefore $-\delta_j V_{\text{max},j}^\circ \leq \hat{v}_j \leq V_{\text{max},j}^\circ$ for v_1 through v_5 , with δ_j and $V_{\text{max},j}^\circ$ as assigned above, and $-1000 \leq \hat{v}_j \leq 1000$ mmol gDW⁻¹ h⁻¹ for the fourteen exchange reactions, wide enough to be effectively unconstrained. With the feasible box fixed, the one remaining ingredient is the objective, here the maximization of urea export. Exchange reaction b_4 represents urea crossing the system boundary and, following the same secretion-positive convention, is written urea $\rightarrow \emptyset$, so that positive $\hat{v}_{b_4}$ denotes secretion and negative $\hat{v}_{b_4}$ denotes uptake. Maximizing urea export is therefore maximizing $\hat{v}_{b_4}$ directly, so the objective coefficient is set to $c_{b_4} = 1$ and every other entry of $\mathbf{c}$ to zero: maximizing $\mathbf{c}^\top \hat{\mathbf{v}} = \hat{v}_{b_4}$ drives urea secretion as high as the constraints allow. Solving Equation (6) by linear programming [6] with these bounds and this objective gave a single, sharply determined answer. The entire enzymatic backbone, v_1 through v_4 , saturated at 118.08 mmol gDW⁻¹ h⁻¹, the capacity imposed by argininosuccinate lyase, and the nitric oxide synthase branch v_5 carried zero flux: with arginine supply capped by the same v_2 bottleneck that limits the rest of the cycle, none of it was left over to spend on the competing NOS reaction once urea export is the sole objective. The optimal urea export was 118.08 mmol gDW⁻¹ h⁻¹, attained at $\hat{v}_{b_4} = 118.08$, and every exchange flux carrying carbon or nitrogen into or out of the cycle, carbamoyl phosphate and aspartate on the uptake side, urea and fumarate on the secretion side, scaled to match it, giving the full optimal flux distribution (right panel, Fig. 1). An independent flux-variability analysis confirmed this optimum is unique rather than one vertex among several tied solutions: minimizing and then maximizing each of the nineteen fluxes in turn, with the objective held at its optimal value, gave an identical minimum and maximum for every reaction to numerical tolerance, so “a single, sharply determined answer” is a checked property of this model rather than an assumption.

The single answer above treats every bound as exact, but each was assembled from database estimates that carry their own uncertainty, and propagating that uncertainty turns the point prediction into a distribution. Assigning each turnover number and the reference abundance a lognormal factor of roughly two, each standard free energy a normal spread of a few kilojoules per mole, and four of the five saturation factors (the fifth, f_2 at argininosuccinate lyase, held fixed at one for lack of a measured substrate) a substrate concentration and Michaelis constant drawn from the compilation of Park and coworkers [21], and re-solving the linear program over ten thousand parameter draws by the procedure of Algorithm 1, a Monte Carlo propagation of assumed parameter distributions rather than a bootstrap resampled from replicate measurements, spreads urea export over

a right-skewed band with median 105 and mean 156 mmol gDW⁻¹ h⁻¹ and a central ninety-five percent interval of 17 to 614. A capacity-only bootstrap, holding every saturation factor at one and letting only the turnover numbers, the reference abundance, and the standard free energies vary, gives a median urea export of 113 mmol gDW⁻¹ h⁻¹, and restoring the four measured saturation factors leaves the median at 105, demonstrating their inertness. The width traces almost entirely to the capacity of argininosuccinate lyase: the reversibility switches never bind, because the cycle runs forward at the optimum, and the four saturation factors that Park and coworkers do constrain, for argininosuccinate synthetase, arginase, ornithine transcarbamylase, and nitric oxide synthase, each leave their reaction with capacity to spare above the v_2 bottleneck, so the sampled bound on v_2 dominates the uncertainty in most draws, and the nitric oxide synthase branch, though it carries a small positive mean and a nonzero spread in the rare draws where it turns on, remains negligible overall.

That the saturation gateway barely moves the prediction is itself the result. Opening f_j with every substrate concentration the database supplies leaves the urea forecast where the capacity scale put it, because the one reaction that sets the answer, argininosuccinate lyase, has no measured substrate: argininosuccinate is a committed cycle intermediate absent from the central-metabolism compilation, so its saturation factor stays at the information-free default while the four measured factors prove inert. Imputing that missing factor from a physiological argininosuccinate concentration and a BRENDA Michaelis constant, and sampling it alongside the rest, drags the urea median down to 51 mmol gDW⁻¹ h⁻¹ and widens the band toward zero (Fig. 2), the shift and the spread both governed by the single quantity the model never measured. A sharper prediction here would come not from more data in general but from data on the reaction that binds.

Algorithm 1 Parametric bootstrap for urea-cycle flux uncertainty

Require: nominal k_{cat}° , e° , ΔG° ; nominal saturation inputs $[S_j]$, $\overline{K_{M,j}}$ where measured; spreads $\sigma_{\ln}$, $\sigma_{\Delta G}$; threshold ΔG^* ; draws N ; seed

- 1: seed the random generator
- 2: **for** $i = 1$ to N **do**
- 3: $e \leftarrow e^\circ \exp(\sigma_{\ln} Z_e)$ ▷ one shared abundance draw, $Z_e \sim \mathcal{N}(0, 1)$
- 4: **for** each enzymatic reaction j **do**
- 5: $k_{\text{cat},j} \leftarrow k_{\text{cat},j}^\circ \exp(\sigma_{\ln} Z_{k,j})$, $\Delta G_j \leftarrow \Delta G_j^\circ + \sigma_{\Delta G} Z_{\Delta G,j}$ ▷ independent
 $Z_{k,j}, Z_{\Delta G,j} \sim \mathcal{N}(0, 1)$ per reaction
- 6: $\delta_j \leftarrow \mathbb{I}[\Delta G_j > \Delta G^*]$
- 7: **if** substrate data measured for reaction j **then**
- 8: $[S_j] \leftarrow [S_j] \exp(\sigma_{\ln} Z_{S,j})$, $K_{M,j} \leftarrow \overline{K_{M,j}} \exp(\sigma_{\ln} Z_{K,j})$ ▷ independent per reaction
- 9: $f_j \leftarrow [S_j] / (K_{M,j} + [S_j])$
- 10: **else**
- 11: $f_j \leftarrow 1$
- 12: **end if**
- 13: $V_{\text{max},j} \leftarrow k_{\text{cat},j} e$; $u_j \leftarrow V_{\text{max},j} f_j$, $\ell_j \leftarrow -\delta_j V_{\text{max},j} f_j$
- 14: **end for**
- 15: $\hat{\mathbf{v}}^{(i)} \leftarrow \arg \max \{ \mathbf{c}^\top \mathbf{v} : \mathbf{S} \mathbf{v} = \mathbf{0}, \ell \leq \mathbf{v} \leq \mathbf{u} \}$
- 16: **end for**
- 17: **return** per-reaction mean, standard deviation, and 2.5/50/97.5 percentiles of $\{ \hat{\mathbf{v}}^{(i)} \}_{i=1}^N$

This result was reached without a single concentration, expression level, or regulatory parameter measured in HL-60 cells: every bound came from a public database, eQuilibrator for the sign of δ_j

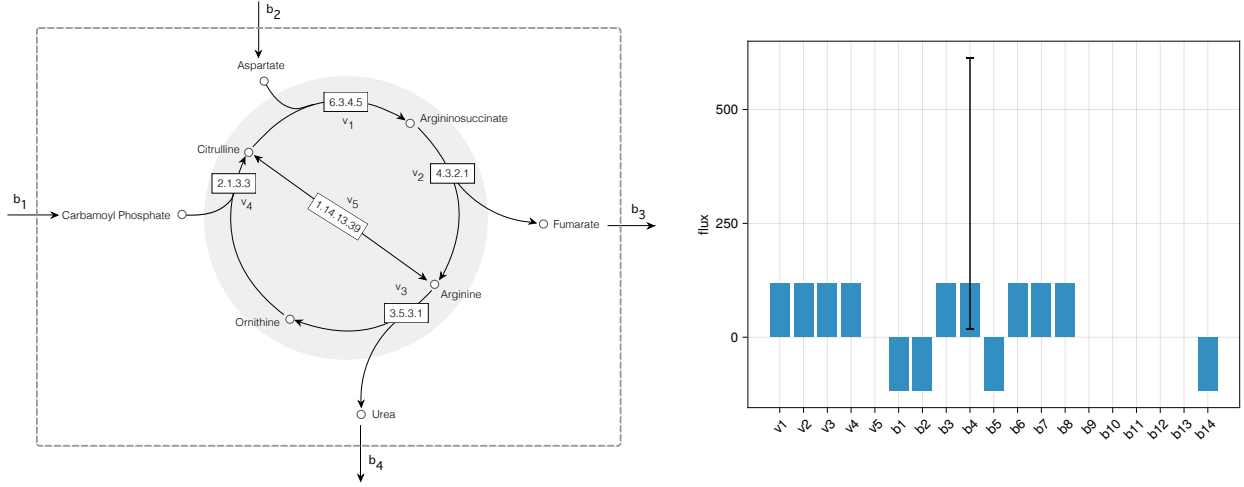


Figure 1: The urea-cycle worked example. Left: the reaction network, with the four cyclic enzymes v_1 through v_4 and the competing nitric oxide synthase branch v_5 labeled by EC number, and the boundary exchanges (carbamoyl phosphate b_1 and aspartate b_2 entering, fumarate b_3 and urea b_4 leaving) crossing the dashed system boundary; arginine, citrulline, ornithine, and argininosuccinate remain internal to the cycle. Right: the optimal flux distribution for the linear program of Equation (6), with bounds set by the thermodynamic and kinetic gateways and the objective $c_{b_4} = 1$ maximizing urea export; bar heights give the flux $\hat{v}_j$ (mmol gDW⁻¹ h⁻¹) at the optimum. Because the cycle is linear, every nonzero backbone flux nearly equals the single throughput up to sign (v_1/v_2 and v_3/v_4 differ from each other only in the rare draws where the nitric oxide synthase branch activates), so the parametric-bootstrap uncertainty of Algorithm 1 is essentially one quantity common to them all; it is drawn as the 2.5–97.5 percentile whisker on the urea-export flux b_4 , the reported objective, rather than repeated identically on every bar. The enzymatic backbone v_1 through v_4 saturates at 118.08, the capacity of argininosuccinate lyase (v_2), while the nitric oxide synthase branch v_5 carries zero flux. Exchange fluxes b_1 through b_{14} follow the secretion-positive convention, so positive bars denote net secretion and negative bars net uptake; b_4 (urea) shows the maximized export flux of 118.08 mmol gDW⁻¹ h⁻¹.

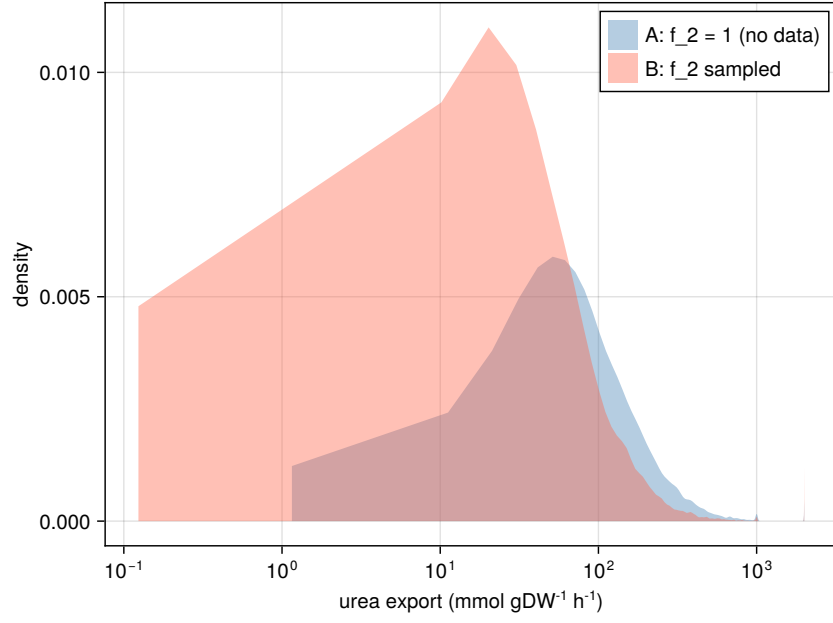


Figure 2: Saturation-gateway sensitivity for urea export. Configuration A (blue) samples the four saturation factors that Park and coworkers [21] constrain and holds $f_2 = 1$ at the argininosuccinate lyase bottleneck, whose substrate is unmeasured, giving a median urea export of $105 \text{ mmol gDW}^{-1} \text{ h}^{-1}$. Configuration B (red) imputes f_2 from a physiological argininosuccinate concentration and a BRENDA Michaelis constant and samples it as well, pulling the median down to $51 \text{ mmol gDW}^{-1} \text{ h}^{-1}$ and widening the band toward zero. The horizontal axis is logarithmic.

and BRENDA for the scale of $V_{\max,j}^\circ$, read against a textbook reference enzyme abundance. Two of the four gateways in Equation (7) were opened, and two were left at their default of unity, and the flux distribution of Figure 1 is the direct consequence: a single hard bottleneck at v_2 and a competing branch at v_5 that remains negligible in typical draws, both explainable entirely from the reversibility and capacity assigned above. Reopening the regulatory gateway θ_j , so that a reaction’s capacity depends on the cell’s own control state rather than sitting fixed at its database-derived ceiling, is the subject of the feedback-inhibition example that follows.

6 Worked Example: A Feedback-Inhibited Linear Pathway

The urea-cycle example opened the thermodynamic and kinetic gateways of Equation (7) and held both the expression ratio (e/e°) and the regulatory factor θ_j at their information-free default of unity. A first pass at feedback exercises θ_j alone: the fast allosteric inhibition in which the accumulating end product binds the committed enzyme and throttles the catalytic activity of the molecules already present, the classical picture of end-product inhibition, with the expression ratio still sitting at unity. A committed biosynthetic step, though, is controlled by its own end product at two levels and on two timescales at once. Alongside the fast allosteric loop the same product represses transcription of the gene encoding the enzyme, a slow loop that lowers the number of enzyme molecules the cell builds rather than the activity of the molecules on hand, and this transcriptional loop is a canonical negative-feedback motif of gene-regulatory circuits, the operon repressed by its own pathway product exactly as the tryptophan operon is repressed by tryptophan. The two loops draw on different currencies, one on catalytic activity and one on enzyme abundance, and they relax on different timescales, so an honest capacity for the committed step has to carry both. The pathway kept for the comparison is the same abstract linear chain that isolates a single control point,



with r_0 the committed uptake step, r_1 and r_2 interconversions, and r_3 the export step, and the end product X_3 now closing both loops onto r_0 at once: a high steady-state X_3 simultaneously inhibits the activity of the enzyme that catalyzes r_0 and represses transcription of its gene. Opening both gateways together on that single step is the whole of the construction.

The truth was generated from a Biochemical Systems Theory S-system, the power-law formalism in which every reaction rate is a product of the participating species raised to real-valued kinetic orders [3, 4], but now the committed enzyme is no longer a fixed parameter. Its transcript m and the enzyme protein E_0 are dynamic species carried alongside the three metabolites X_1 , X_2 , and X_3 , five state variables integrated together to a single self-consistent steady state with BSTModelKit.jl [22]. The end product enters the kinetics twice. In the rate of the committed step it appears with a negative kinetic order $-a$, the allosteric activity loop, so that a higher X_3 lowers the throughput the enzyme molecules already present can sustain; in the rate of transcription of the enzyme’s gene it appears with a second negative kinetic order $-b$, the expression loop, so that a higher X_3 lowers the rate at which new message, and with it new enzyme, is made. Each power law is the concrete operating-point instance of the partition-function control of Equations (17) and (18) [14]: the Boltzmann sum over regulatory microstates, evaluated near the working point of a single corepressing effector, collapses to a power of that effector, the reduction that gives both the allosteric term θ_{true} and the transcriptional repression their power-law form. The two families of balance follow the convention established earlier, the three metabolite balances enforcing $\mathbf{S}\hat{\mathbf{v}} = \mathbf{0}$ with no growth-dilution term while the growth rate μ lives in the transcript and protein balances of

Equations (14) and (15), where an intracellular macromolecule is cleared both by its own turnover and by dilution into the growing culture. The committed-step rate carries both controls,

$$v_{r_0} = V_{\max, r_0}^{\circ} (e/e^{\circ}) \theta_{\text{true}}, \quad \theta_{\text{true}} = X_3^{-a}, \quad a = 0.4, \quad \text{transcription} \propto X_3^{-b}, \quad b = 0.6. \quad (21)$$

The expression ratio (e/e°) was not read from a table but computed from the sequences the enzyme is made from. The transcription and translation capacities r_X and r_L follow from a 330-residue enzyme and its 990-nucleotide gene, the polymerase elongating at 60 nucleotides per second and the ribosome at 16 amino acids per second; together with an mRNA half-life of 2.5 minutes, a protein half-life of 35 minutes, and a 60-minute culture doubling time, these fix the residence times of message and enzyme in the steady-state balances of Equations (14) and (15), and with them the reference enzyme abundance e° against which Equation (16) normalizes the ratio (e/e°) , the response of that ratio to the end product being carried by the repression order b rather than by the residence times, which cancel from it. The growth rate enters those balances in a way that repays stating precisely, because it is easy to misread. There is one μ , the specific growth rate of the culture, and it is identical in both denominators, $\theta_m + \mu$ for the transcript and $\theta_p + \mu$ for the enzyme; what differs between the two is never μ but only each species' own first-order turnover θ . The consequence is a clean signature. The transcript turns over fast, so that same μ is only about four percent of its total clearance $\theta_m + \mu \approx 0.289 \text{ min}^{-1}$; the enzyme is stable, so the identical μ is more than a third, about thirty-seven percent, of its total clearance $\theta_p + \mu \approx 0.031 \text{ min}^{-1}$. Growth dilution is therefore a dominant clearance term for the enzyme and a negligible one for its message, though the number diluting them is one and the same, and it is exactly this asymmetry that justifies carrying μ in every intracellular balance rather than folding it into a degradation constant.

Because the chain is linear, conservation forces every metabolic reaction to carry one common throughput T , and because the export step r_3 is first order in X_3 , that throughput pins the end-product level at $X_3 = T/k_3$. Substituting the two controls back into the committed-step rate, the truth model's own allosteric term $\theta_{\text{true}} = X_3^{-a}$ and the expression ratio the transcriptional repression sets through Equation (16), reduces the coupled five-state system to a single self-consistency condition in X_3 alone, and the value that condition selects is the value the full numerical integration confirmed: $X_3^* \approx 4$, an allosteric term $\theta_{\text{true}}^* \approx 0.574$, an expression ratio $(e/e^{\circ})^* \approx 0.464$, and a throughput $T^* \approx 2.66$, less than a third of the uninhibited capacity of the committed step (Fig. 3). The control scheme behind those numbers is the standard negative one: a repressor that X_3 corepresses binds the operator of the enzyme's gene and throttles transcription downward as the end product accumulates, but only down to a basal leak floor, the residual expression λ set by the basal occupancy $u^{\dagger}$ of Equations (14) and (18), so that the gene is never silenced completely and a trickle of enzyme is always made. The two feedback strengths were chosen so that each gateway is individually load-bearing, $a = 0.4$ on the activity loop and $b = 0.6$ on the expression loop: neither factor alone accounts for the observed throttling, and a model that opened one and defaulted the other would miss the mark by a visible margin.

The flux-balance model kept the chain's stoichiometry unchanged and placed the entire dual-control content on the single bound of the committed step,

$$0 \leq \hat{v}_{r_0} \leq V_{\max, r_0}^{\circ} (e/e^{\circ}) \theta_{\text{FBA}}(X_3), \quad V_{\max, r_0}^{\circ} = 10, \quad (22)$$

the expression and regulatory instance of the general bound of Equation (7), with the thermodynamic switch and the saturation factor left at their defaults and the two remaining gateways, expression (e/e°) and activity $\theta_{\text{FBA}}(X_3)$, both in play. The activity gateway does not reuse the

truth model's own kinetic exponent θ_{true} from Equation (21); a bound built from the exact functional form that generates the quantity it is meant to predict would make the recovery an identity rather than a check. Instead θ_{FBA} is a bounded, two-state Hill occupancy,

$$\theta_{\text{FBA}}(X_3) = \frac{K^n}{K^n + X_3^n}, \quad K = 5, \quad n = 2, \quad (23)$$

the same partition-function-motivated form used for f_s in Equation (17), with K and n chosen independently of the truth model's allosteric order a rather than fit to reproduce it, and with no basal-leak floor: unlike the transcriptional loop, a direct allosteric on/off switch has no leaky analog, so $\theta_{\text{FBA}} \rightarrow 0$ as $X_3 \rightarrow \infty$. Both gateway factors were evaluated from the measured steady state, the enzyme abundance and the end-product level read off the integrated model, and not copied from the integrated flux the linear program is meant to reproduce. Four solves of the linear program of Equation (6) differed in nothing but this one bound. A feedback-blind solve, holding both factors at unity, overshoot to the uninhibited capacity of 10, roughly 3.8 times the true throughput; opening the expression gateway alone throttled the committed step to 4.64; opening the activity gateway alone throttled it to 6.10; and only opening both together landed on 2.83, approximating the BST truth of 2.66 to within about six percent (Fig. 3). The escalation is the whole point. Nothing about either feedback loop appears anywhere in the stoichiometric matrix, which is identical whether X_3 closes a single loop, both loops, or none; the entire dual-control fact lives in the one bound on r_0 , where the two gateways enter as independent multiplicative factors and each removes a distinct, quantifiable slice of the capacity. Read as a ledger, the uninhibited capacity of 10 was marked down once by expression to 4.64 and once by activity to 6.10, and only the product of the two markdowns approached the throughput the cell actually ran.

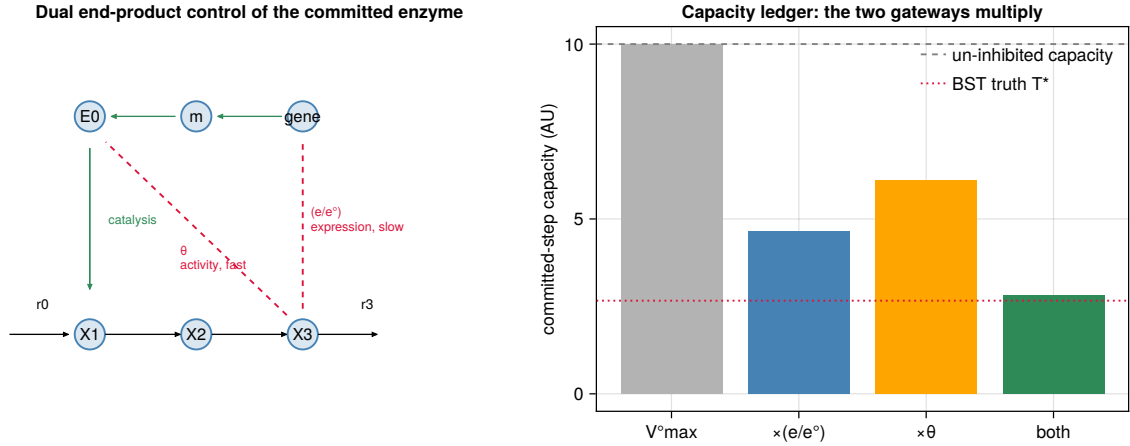


Figure 3: The dual-control feedback worked example. Left: the wiring of the linear chain of Equation (20), in which the end product X_3 closes two negative loops onto the committed step r_0 , repressing transcription of the enzyme's gene, the slow expression loop acting through (e/e°) , and inhibiting the activity of the enzyme E_0 already present, the fast allosteric loop acting through θ_{FBA} . Right: the capacity ledger for the committed-step bound of Equation (22). The uninhibited capacity $V^\circ_{\text{max}, r_0} = 10$ is reduced by the expression gateway alone to 4.64 and by the activity gateway alone to 6.10, and only the two gateways together approach the throughput of the Biochemical Systems Theory truth, $T^* \approx 2.66$, landing at 2.83; the dashed line marks the uninhibited capacity.

7 Integration in Practice

The urea-cycle and feedback examples opened the gateways of Equation (7) one at a time, on networks small enough that a hand calculation could check the result against a known truth: one degree of freedom in each case, $n - m = 1$, far short of the large right null space that motivates the gateway framework at genome scale (Section 3). Two reported systems show the same gateways opened together, at a scale no hand calculation could reach, and driven by measurement rather than by a modeler’s guess at what a bound should be. The first couples metabolism and regulation into one dynamic model of a cell-free reaction; the second uses the same bounds not to estimate a flux distribution from data but to design one.

Vilkhovoy and coworkers built the metabolic half of that coupling on a sequence-specific reconstruction of *E. coli* cell-free protein synthesis, in which the transcription and translation reactions required to make a target protein from its loaded DNA and mRNA sequence are computed directly from those sequences and entered into the stoichiometric matrix as flux-bearing chemistry alongside metabolism itself [20], the sequence-to-stoichiometry construction described earlier and credited there as the mechanism behind this capstone. Control over that expression machinery, the fraction of promoter and ribosome capacity actually switched on at a given moment, is supplied by the effective biophysical partition-function model of Adhikari and coworkers [14], the same statistical-mechanical control function of Equations (17) and (18), built earlier from a Boltzmann sum over regulatory microstates and worked by hand for a single feedback loop. In this setting the same θ_j machinery is asked to populate every regulated step of a larger network at once, rather than one control point on one pathway. They then closed the loop dynamically [8]: metabolism, expression, and regulation are integrated into a single constraint-based model whose bounds are recomputed as the reaction proceeds, so that the thermodynamic, kinetic, expression, and regulatory gateways are opened together rather than in isolation. This integration is checked against denser data: time-resolved measurement of the intracellular state over the course of a batch reaction, 63 metabolites tracked through the reaction, together with time courses of mRNA, protein abundance, and enzyme activity. Every worked example earlier in this chapter compared a flux prediction to a single steady-state target; here the comparison runs against a trajectory, resolved in both time and in the identity of the species tracked, at a resolution the small hand-checked networks never had reason to offer.

This density of measurement is what makes cell-free protein synthesis the natural home for the term that ordinary flux balance analysis sets aside. A cell-free reaction has no growing culture to dilute into, so the specific growth rate μ is zero, but the reaction is manifestly not at steady state: transcript, protein, and metabolite pools rise and fall over the course of the batch, so the pseudo-steady-state assumption that collapses Equation (1) to Equation (4) fails from the direction opposite to slow growth. The accumulation that assumption would otherwise discard is exactly what the sixty-odd tracked metabolites record, and because the reaction vessel and the intracellular compartment are one and the same volume, with no membrane separating a sampled extracellular phase from an inferred intracellular one, that accumulation is read directly off the measurement rather than reconstructed indirectly from an uptake or secretion rate. This is the setting pointed to when Equation (5) was first written down: cell-free integration is where the growth-dilution and accumulation term Equation (5) discards is, for once, under direct experimental observation rather than assumed away.

The second capstone runs the same construction in the opposite direction. Wayman and coworkers grafted the N-linked glycosylation pathway of *Campylobacter jejuni*, the source of the designer glycans used to improve the pharmacokinetic and immunogenic properties of therapeutic glycoproteins, onto a genome-scale reconstruction of *E. coli* metabolism, and then searched the combined

network for gene deletions predicted to couple glycan output to growth itself, so that a strain could not grow well without also making the glycan [9]. Where the cell-free capstone read the gateways of Equation (7) from measurement to obtain a flux estimate, this campaign set them directly: a reaction knocked out has its bound collapsed to zero, the capacity gateway of Equation (7) closed rather than left open, and each candidate deletion, or combination of deletions, defines a distinct box $[\ell, \mathbf{u}]$ under which the linear program is re-solved for the resulting flux distribution, entirely inside the model and before a single strain is built. Knocking out a reaction is the simplest way to reach into a bound; the same gateway can in principle be narrowed by attenuating an enzyme’s expression or widened by over-expressing it, moving the (e/e°) factor of Equation (7) down or up rather than collapsing it to zero. The deletions this campaign screened were sufficient on their own: the best single knockout the model identified, chosen for its predicted growth-coupling among the candidate deletions, raised glycan production roughly threefold over the unmodified strain once built and measured; the linear program re-solved under each candidate’s modified bound predicted which deletion to make, not the magnitude of the yield increase eventually observed. The bounds are, in this use, not a report of what the cell is doing but a lever for what the cell is made to do.

Read together, the two capstones run the same operation on Equation (7) in opposite directions, once the intervention itself has been chosen. In the cell-free system, measurement narrows the gateways and the flux distribution is estimated; in the glycan strain, the engineer narrows a gateway and the flux distribution is designed. Once a set of bound changes is specified, both problems solve the identical linear program under the modified box, since fixing ℓ and $\mathbf{u}$ and re-solving the same constrained optimization is all either use requires; only the direction in which information moves through the gateway, from data inward for estimation, from intention outward for design, changes. Choosing which bounds to change is not itself part of that shared operation, though: for design, searching a combinatorial space of candidate deletions or expression changes for one that couples the desired product to growth is a discrete optimization problem in its own right, solved by frameworks such as bilevel programming [10] rather than by the continuous linear program each candidate is then checked against. Estimation and design are, at bottom, the same evaluation of the bounds, run in two directions; only design additionally requires searching for which bounds to move.

8 Outlook

The worked examples of this chapter populated the gateways of Equation (7) from tabulated sources: standard free energies from group-contribution estimates, turnover numbers from enzyme databases, a control function built by hand from a partition function over a handful of regulatory microstates. None of that is required by the construction itself. A turnover number, a saturation constant, or a control function θ_j is, mathematically, a parameter or a function fit to data, and where time-resolved measurement of the kind reviewed in the capstones is available, these quantities can be learned directly from the data they are meant to explain rather than looked up from a compilation assembled under different conditions, with the tabulated values retained as a prior where dynamic measurement is not.

The same construction, opened here on networks small enough to check by hand, carries without modification to genome-scale reconstructions of thousands of reactions: Equation (7) is applied reaction by reaction regardless of network size. What changes with scale is the weight the bounds are asked to carry. The null space of $\mathbf{S}$ grows with the excess of reactions over metabolites, so that at genome scale the stoichiometric constraint alone leaves an enormous subspace of flux vectors indistinguishable, and it is precisely there that a poorly constrained gateway, a bound left at its information-free default for want of data, does the most damage to a prediction. Closing gateways

with data matters most exactly where the null-space argument bites hardest.

The dynamic, data-rich regime realized in the cell-free capstone is the natural home for the growth-dilution and accumulation term retained in Equation (4) and dropped in Equation (5). Cell-free reactions are one setting dense enough in time-resolved measurement to estimate fluxes without discarding that term, and the same opportunity extends wherever intracellular state can be tracked closely enough in time, in principle inside a growing culture as readily as in a reaction tube. The natural direction is therefore not a retreat to the steady-state reduction of Equation (5) once more data arrives, but the opposite: estimating flux directly against Equation (4), with $\mu\mathbf{x}$ carried rather than discarded, wherever the measurement to support it exists.

References

- [1] Aarash Bordbar, Jonathan M. Monk, Zachary A. King, and Bernhard O. Palsson. Constraint-based models predict metabolic and associated cellular functions. *Nature Reviews Genetics*, 15(2):107–120, 2014. doi: 10.1038/nrg3643.
- [2] Leonor Michaelis and Maud L. Menten. Die kinetik der invertinwirkung. *Biochemische Zeitschrift*, 49:333–369, 1913.
- [3] Michael A. Savageau. *Biochemical Systems Analysis: A Study of Function and Design in Molecular Biology*. Addison-Wesley, Reading, MA, 1976.
- [4] Michael A. Savageau, Eberhard O. Voit, and Douglas H. Irvine. Biochemical systems theory and metabolic control theory: 1. Fundamental similarities and differences. *Mathematical Biosciences*, 86(2):127–145, 1987. doi: 10.1016/0025-5564(87)90007-1.
- [5] Timothy E. Allen and Bernhard Ø. Palsson. Sequence-based analysis of metabolic demands for protein synthesis in prokaryotes. *Journal of Theoretical Biology*, 220(1):1–18, 2003.
- [6] Jeffrey D. Orth, Ines Thiele, and Bernhard Ø. Palsson. What is flux balance analysis? *Nature Biotechnology*, 28(3):245–248, 2010. doi: 10.1038/nbt.1614.
- [7] Laurent Heirendt, Sylvain Arreckx, Thomas Pfau, Sebastian N. Mendoza, Anne Richelle, Almut Heinken, Hulda S. Haraldsdóttir, Jacek Wachowiak, Sarah M. Keating, Vanja Vlasov, Sigrid Magnusdóttir, Chiam Yu Ng, German Preciat, Alise Žagare, Siu H. J. Chan, Maike K. Aurich, Catherine M. Clancy, Jennifer Modamio, John T. Earls, David S. Wishart, and Ronan M. T. Fleming. Creation and analysis of biochemical constraint-based models: the COBRA toolbox v3.0. *Nature Protocols*, 14(3):639–702, 2019. doi: 10.1038/s41596-018-0098-2.
- [8] M. Vilkhovoy, S. Dammalapati, S. Vadhin, A. Adhikari, and J. Varner. Integrated constraint-based modeling of E. coli cell-free protein synthesis. *bioRxiv*, 2023. doi: 10.1101/2023.02.10.528035. preprint.
- [9] Joseph A. Wayman, Cameron Glasscock, Thomas J. Mansell, Matthew P. DeLisa, and Jeffrey D. Varner. Improving designer glycan production in Escherichia coli through model-guided metabolic engineering. *Metabolic Engineering Communications*, 9:e00088, 2019. doi: 10.1016/j.mec.2019.e00088.
- [10] Anthony P. Burgard, Priti Pharkya, and Costas D. Maranas. OptKnock: A bilevel programming framework for identifying gene knockout strategies for microbial strain optimization. *Biotechnology and Bioengineering*, 84(6):647–657, 2003. doi: 10.1002/bit.10803.

- [11] Moritz E. Beber, Mattia G. Gollub, Dana Mozaffari, Kevin M. Shebek, Avi I. Flamholz, Ron Milo, and Elad Noor. eQuilibrator 3.0: a database solution for thermodynamic constant estimation. *Nucleic Acids Research*, 50(D1):D603–D609, 2022. doi: 10.1093/nar/gkab1106.
- [12] Antje Chang, Lisa Jeske, Sandra Ulbrich, Julia Hofmann, Julia Koblitiz, Ida Schomburg, Meina Neumann-Schaal, Dieter Jahn, and Dietmar Schomburg. BRENDA, the ELIXIR core data resource in 2021: new developments and updates. *Nucleic Acids Research*, 49(D1):D498–D508, 2021. doi: 10.1093/nar/gkaa1025.
- [13] Ron Milo, Paul Jorgensen, Uri Moran, Griffin Weber, and Michael Springer. BioNumbers—the database of key numbers in molecular and cell biology. *Nucleic Acids Research*, 38:D750–D753, 2010. doi: 10.1093/nar/gkp889.
- [14] Abhinav Adhikari, Michael Vilkhovoy, Sandra Vadhin, Ha Eun Lim, and Jeffrey D. Varner. Effective biophysical modeling of cell free transcription and translation processes. *Frontiers in Bioengineering and Biotechnology*, 8:539081, 2020. doi: 10.3389/fbioe.2020.539081.
- [15] Christopher S. Henry, Linda J. Broadbelt, and Vassily Hatzimanikatis. Thermodynamics-based metabolic flux analysis. *Biophysical Journal*, 92(5):1792–1805, 2007. doi: 10.1529/biophysj.106.093138.
- [16] Benjamín J. Sánchez, Cheng Zhang, Avlant Nilsson, Petri-Jaan Lahtvee, Eduard J. Kerkhoven, and Jens Nielsen. Improving the phenotype predictions of a yeast genome-scale metabolic model by incorporating enzymatic constraints. *Molecular Systems Biology*, 13(8):935, 2017. doi: 10.15252/msb.20167411.
- [17] Edward J. O’Brien, Joshua A. Lerman, Roger L. Chang, Daniel R. Hyduke, and Bernhard Ø. Palsson. Genome-scale models of metabolism and gene expression extend and refine growth phenotype prediction. *Molecular Systems Biology*, 9:693, 2013. doi: 10.1038/msb.2013.52.
- [18] Markus W. Covert, Christophe H. Schilling, and Bernhard Palsson. Regulation of gene expression in flux balance models of metabolism. *Journal of Theoretical Biology*, 213(1):73–88, 2001. doi: 10.1006/jtbi.2001.2405.
- [19] Sriram Chandrasekaran and Nathan D. Price. Probabilistic integrative modeling of genome-scale metabolic and regulatory networks in *Escherichia coli* and *Mycobacterium tuberculosis*. *Proceedings of the National Academy of Sciences*, 107(41):17845–17850, 2010. doi: 10.1073/pnas.1005139107.
- [20] Michael Vilkhovoy, Nicholas Horvath, Che-Hsiao Shih, Joseph A. Wayman, Kara Calhoun, James Swartz, and Jeffrey D. Varner. Sequence specific modeling of *E. coli* cell-free protein synthesis. *ACS Synthetic Biology*, 7(8):1844–1857, 2018. doi: 10.1021/acssynbio.7b00465.
- [21] Junyoung O. Park, Joshua D. Rabinowitz, et al. Metabolite concentrations, fluxes and free energies imply efficient enzyme usage. *Nature Chemical Biology*, 12(7):482–489, 2016. doi: 10.1038/nchembio.2077.
- [22] Sandra Vadhin and Jeffrey D. Varner. BSTModelKit.jl: A Julia package for constructing, solving, and analyzing biochemical systems theory models, 2026.